Computational Resources & Reproducibility

Resource	Specification	Note
GPU	NVIDIA GeForce RTX 3070 / 4060	8 GB VRAM, consumer-grade
CPU	AMD Ryzen / Intel Core i7	8–16 cores, any modern CPU
RAM	16–32 GB	Dataset fits entirely in memory
PyTorch Version	2.7.0 + CUDA 11.8	Standard deep learning stack
Training Time (Zone 1)	~20 min (20 epochs)	3,523 windows, batch=64
Training Time (7-farm)	~2.1 h (40 epochs)	27,552 windows, batch=48
Training Time (17-site)	~5.8 h (40 epochs)	62,782 windows, batch=48
Inference Speed	~5 ms / window	200 windows/sec on single GPU
Disk (code + checkpoints)	~250 MB	Compact deployment footprint
Total Experiments	~40 runs (all ablations)	Reproducible in ~48 GPU-hours
Random Seeds	{42, 123, 2024}	Three seeds for variance estimation
Code License	**MIT (Open Source)**	github.com/AL1325651958/LNN_Mamba_2

All experiments reproducible on a single consumer GPU. Full ablation suite completes in ~48 GPU-hours on 8GB VRAM.